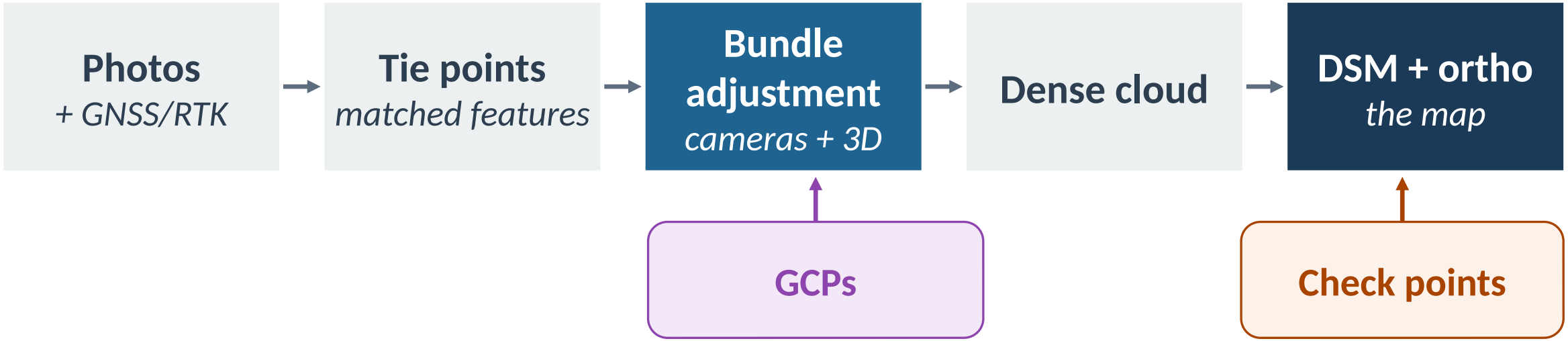
Drone mapping

Ground control, photos for structure from motion, mission planning



From photos to a map



- **Today:** the input – control points, photos, flight plan
- **After the flights:** WebODM, orthomosaic, DSM, accuracy

Wikipedia: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Today

- **Control:** ground control points and check points, targets
- **Photos:** overlap, pixel size, sharpness, light
- **Plan:** from flying height to flight lines and flight time
- **Then, on the field day:** panels out, RTK, flights



1 Control

Ground control points and check points

Photos → map | **Control** | Photos | Plan



Ground control and check points

Ground control point (GCP)

- Known coordinates, visible in the photos
- **Used** in the bundle adjustment
- Fixes position, scale, rotation, height

Check point

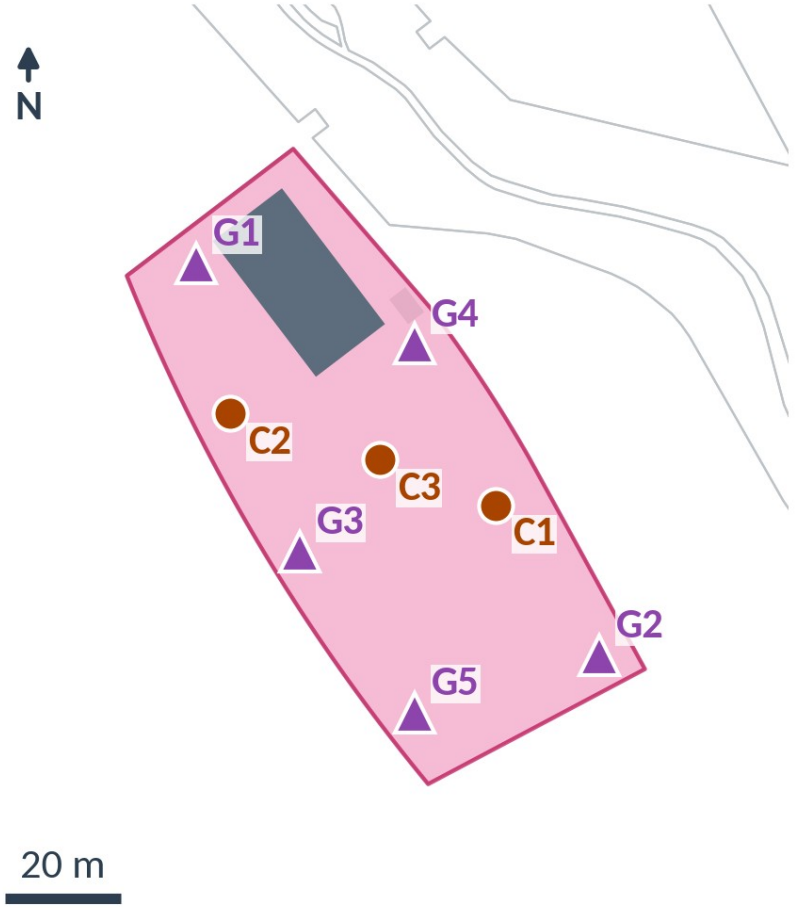
- Measured the same way
- **Not used** in the processing
- Tests the result: error in E, N, H

Pause: why can a GCP not tell you how good your map is?

Wikipedia: Ground control point · Accuracy and precision · Root mean square deviation



How many, and where?



- **≥ 5 GCPs**: edges and middle
- Each one in ≥ 5 photos
- **Check points** in between, never used as GCPs
- Open sky, flat ground, away from walls
- Final places: decided in the field

Wikipedia: Ground control point · Bundle adjustment

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; layout: R. Becker; rules: OpenDroneMap documentation (GCP)

Georeferencing: two questions

corrections from ↓	precise points on the ground	precise camera positions
SAPOS	GCPs + RTK (Emlid) **cm**	drone RTK (mobile data) or PPK **cm**
own base (Emlid)	GCPs, base + rover **cm relative***	drone RTK or PPK from own base **cm relative***
nothing / maps	virtual GCPs: DOP + DGM1 **dm**	drone GNSS only **m**

Every cell: check points · * absolute only if the base position is known

Wikipedia: Real-time kinematic positioning · Ground control point · Orthophoto



Think first

What must my map fit?

my map must …	method	fit
count trees, area of a field	?	
crop change since the last flight	?	
overlay with DOP, ALS, cadastre	?	
volume of a pile, accuracy study	?	

Pause: pick a method from the grid for each task – and why

Wikipedia: Image registration · Accuracy and precision



Solution

What must my map fit?

my map must …	method	fit
count trees, area of a field	drone GNSS only	m
crop change since the last flight	co-register to the first flight	relative
overlay with DOP, ALS, cadastre	virtual GCPs or drone RTK	dm
volume of a pile, accuracy study	RTK/PPK + GCPs + check points	cm

Cheapest method that meets the purpose – proven with check points

Wikipedia: Image registration · Accuracy and precision



Our choice at the Green Fab Lab

- **Ground:** 5 GCPs + 3 check points: SAPOS and our own base
- **Camera:** one mission, three times: SAPOS / own base / none + PPK
- **After the flights:** each with and without GCPs – errors on the check points
- **You find out yourselves which cell is worth the effort**

Wikipedia: Real-time kinematic positioning



Corrections and heights: what you need from geo0260

- **Base on a known point → corrections:** DGNSS dm, RTK cm (FIX)
- **SAPOS HEPS:** network RTK, needs mobile data
- **Own base:** LoRa to the rover, NTRIP to the drone, or PPK
- **Heights:** GNSS h ≈ NHN + 45 m here
- **One height system** for photos, GCPs, check points

Details: slides geo0260 GNSS, RTK and SAPOS (parts 2–5)



Which method when?

	SAPOS + mobile data	SAPOS, dead spot	no network, mobile data	no network, no mobile data
SAPOS network RTK	standard	–	–	–
own base, known mark	rarely	rover by LoRa, drone PPK	standard [1]	rover by LoRa, drone PPK [1]
PPK (drone + base log)	fallback	standard	fallback [1]	standard [1]
drone GNSS only	m-level tasks	m-level tasks	m-level tasks	m-level tasks

[1] mark measured once: long static log + PPP

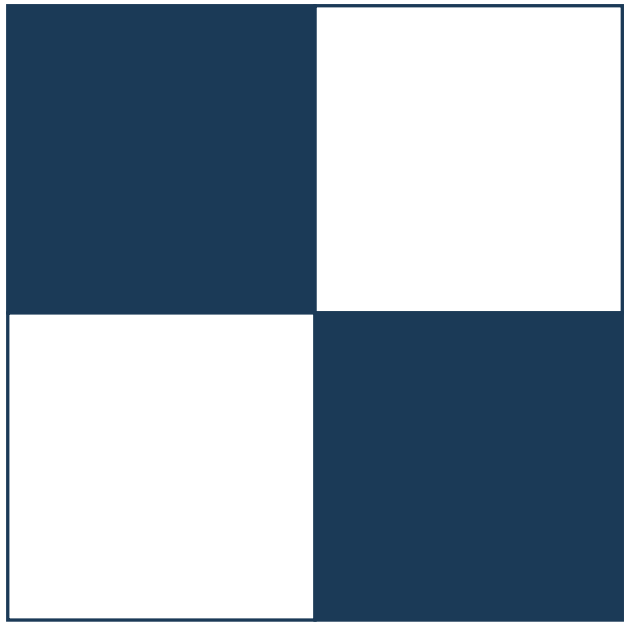
Always: check points



Targets you can find in the photos

50 cm panel	RGB	MS
at 25 m	73 px	44 px
at 60 m	31 px	18 px

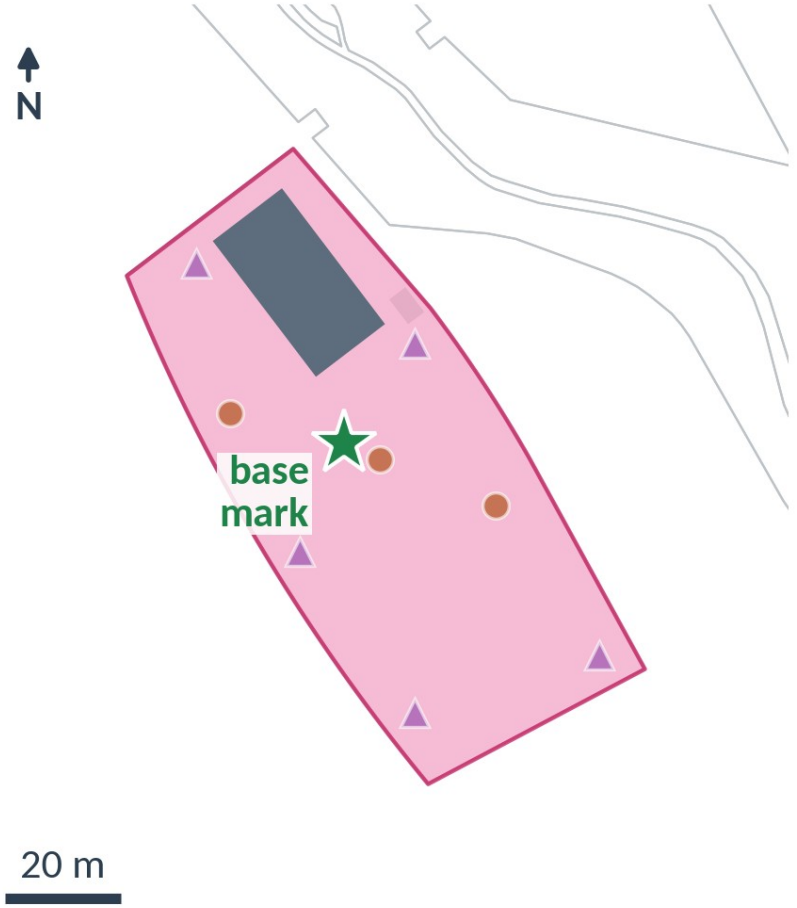
- Measure the centre: FIX, ≈ 30 s
- Measure again later
- Photo of each panel + name
- Leave it until the drone has landed



matte, flat, weighted

Wikipedia: Fiducial marker · Ground sample distance

Our base mark at the Green Fab Lab



- **No official mark nearby** (all destroyed) → our own
- Nail in a paving joint, south of the building
- **Measure:** SAPOS RTK, 2×, ≥ 1 h apart
- **Or check:** 2 h static log + PPK
- Photo, sketch, E N h H
- **Field day:** set it, measure it, base on it
- **Then:** rover and drone with our own corrections

Wikipedia: Survey marker · Tribrach

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; mark status: TIM-online; base mark: planned (R. Becker)

GCPs for WebODM

```
EPSG:25832
329851.42 5708012.37 26.91 2734 1880 DJI_0123.JPG G1
329851.42 5708012.37 26.91 1022 2410 DJI_0124.JPG G1
329893.10 5707990.85 27.05 3911  760 DJI_0157.JPG G2
```

- **Per line:** E N H, pixel x y, photo, name
- Pixels: click in the WebODM GCP interface
- **One height system:** NHN vs. ellipsoid ≈ 45 m
- **Photo heights:** h or EGM96? Check over the base mark

Format: OpenDroneMap documentation, Ground Control Points (docs.opendronemap.org/gcp)



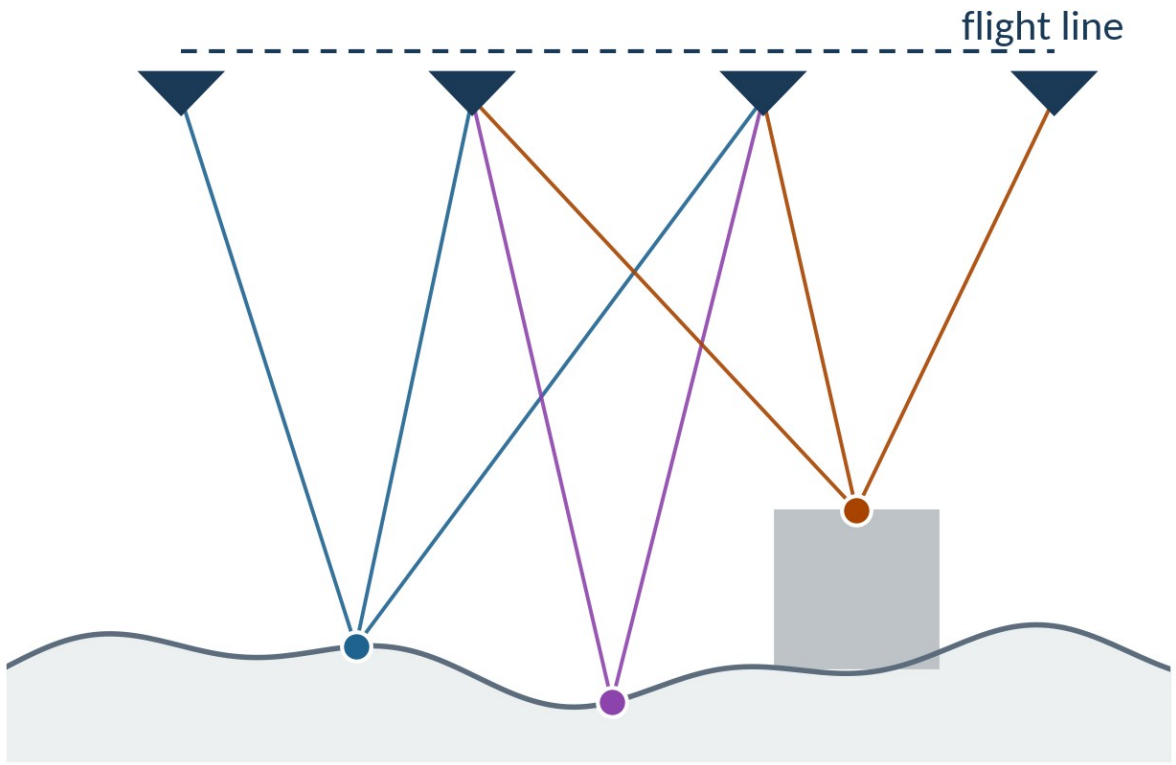
2 Photos

What structure from motion needs

Photos → map | Control | **Photos** | Plan



Structure from motion

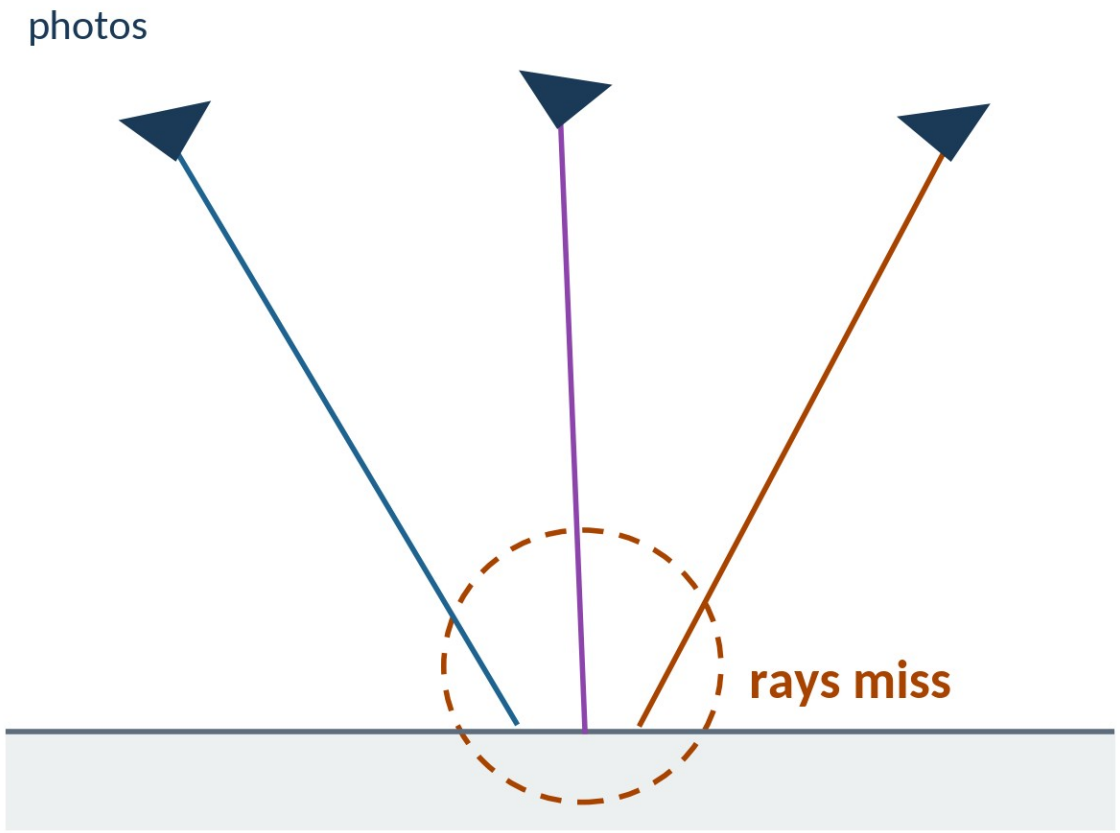


- Same feature in ≥ 3 photos (better 5+)
- Needs texture
- Scene must stand still
- Walls only from oblique photos

Wikipedia: Structure from motion · Feature (computer vision) · Bundle adjustment

Think first

Bundle adjustment: make the rays meet



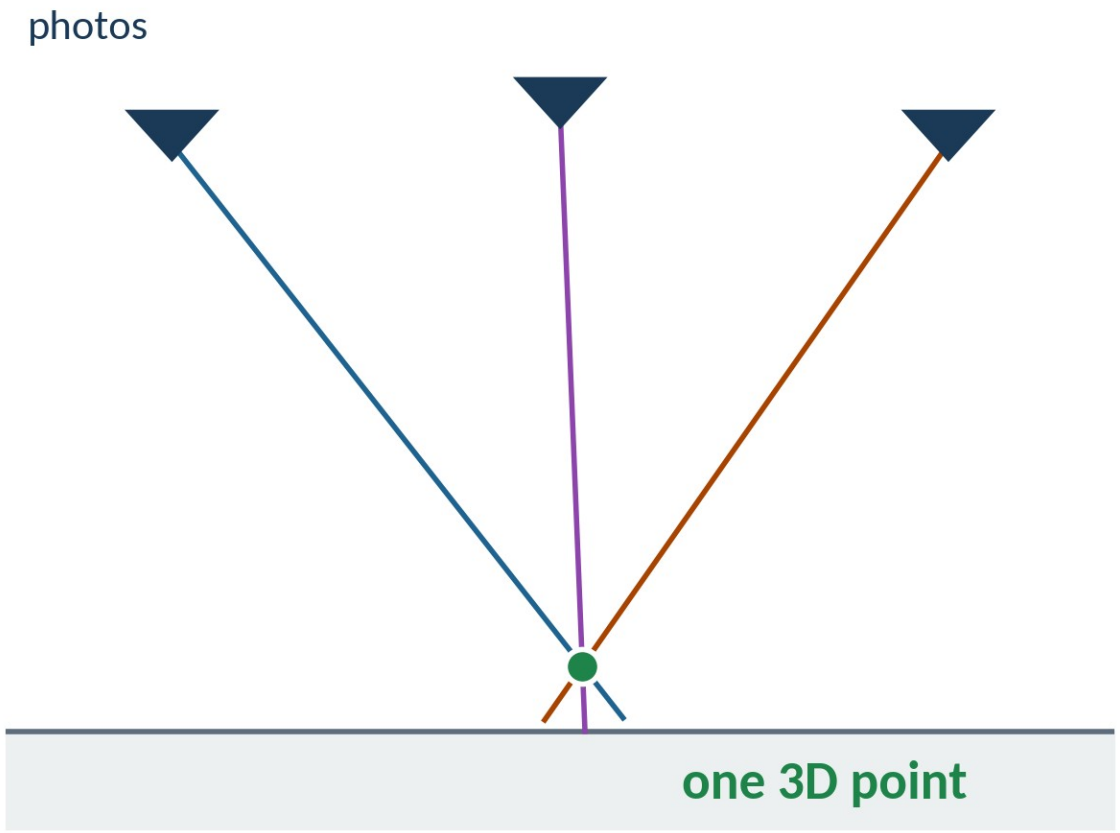
- Each photo: a bundle of rays to its tie points
- First guess (drone GNSS): rays of one point miss

Pause: what can the software change so that the rays meet?

Wikipedia: Bundle adjustment · Triangulation (computer vision)

Solution

Bundle adjustment: make the rays meet



- Each photo: a bundle of rays to its tie points
- First guess (drone GNSS): rays of one point miss

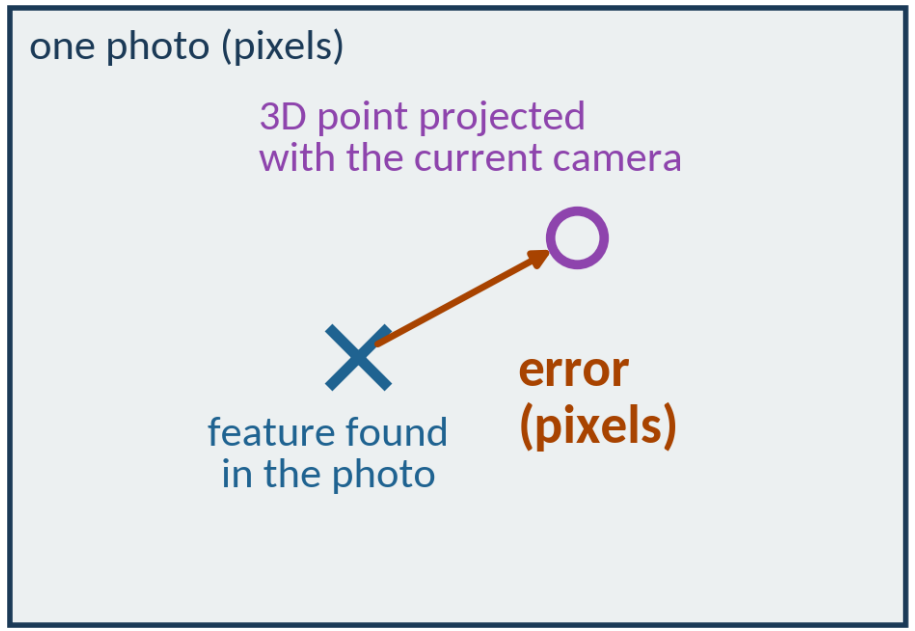
Change all at once:

- camera positions and angles (6 per photo)
- the 3D points (3 each)
- the lens model (focal length, distortion)

least squares over all photos

Wikipedia: Bundle adjustment · Triangulation (computer vision)

Reprojection error



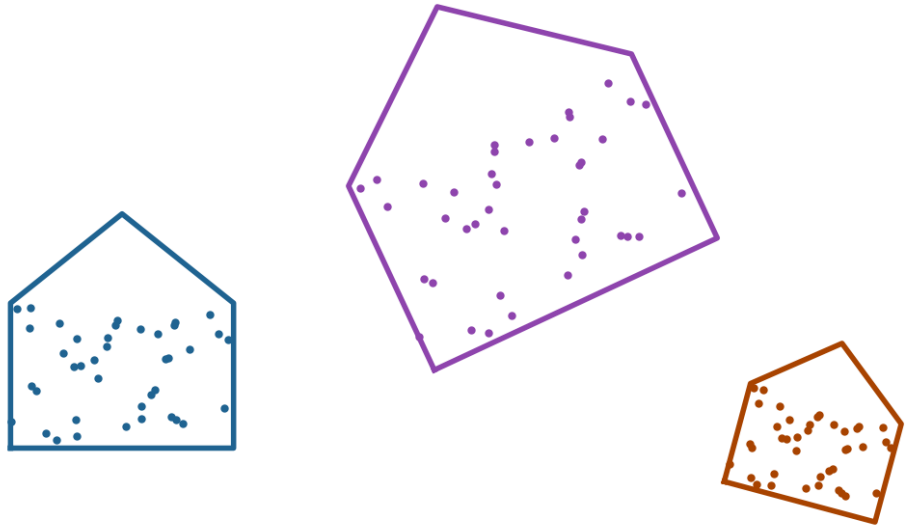
- **Minimise:** Σ (error in pixels)²
- over all points and all photos
- Green Fab Lab, RGB: 131 photos × 6 + thousands of points × 3
- **Report:** mean error < 1 px is good

Wikipedia: Reprojection error · Levenberg–Marquardt algorithm

Think first

A model from photos alone floats

where? how big? which way up?

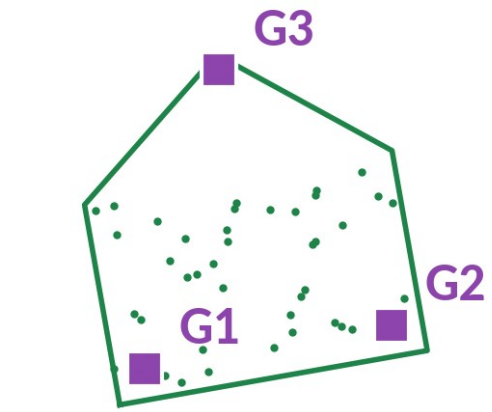


- Photos fix the shape of the model
- No GNSS, no control points

Pause: what can the photos alone not tell? how many numbers are missing?

Wikipedia: Similarity (geometry) · Aerial triangulation

A model from photos alone floats



- Photos fix the shape of the model
- No GNSS, no control points

7 numbers: 3 shift, 3 rotation, 1 scale

Fixed by:

- GCPs (≥ 3, better 5): cm
- camera positions: GNSS m, RTK/PPK cm

Check points stay out!

Wikipedia: Similarity (geometry) · Aerial triangulation

SfM, SLAM, multiple view geometry

Self-study

	multiple view geometry	SfM	SLAM
what	the mathematics	3D from a photo set	where am I + map
when	–	offline, after the flight	online, in real time
bundle adjustment	the method	global, all photos	local, recent frames
example	Hartley & Zisserman	OpenSfM in WebODM	robot, drone hovering

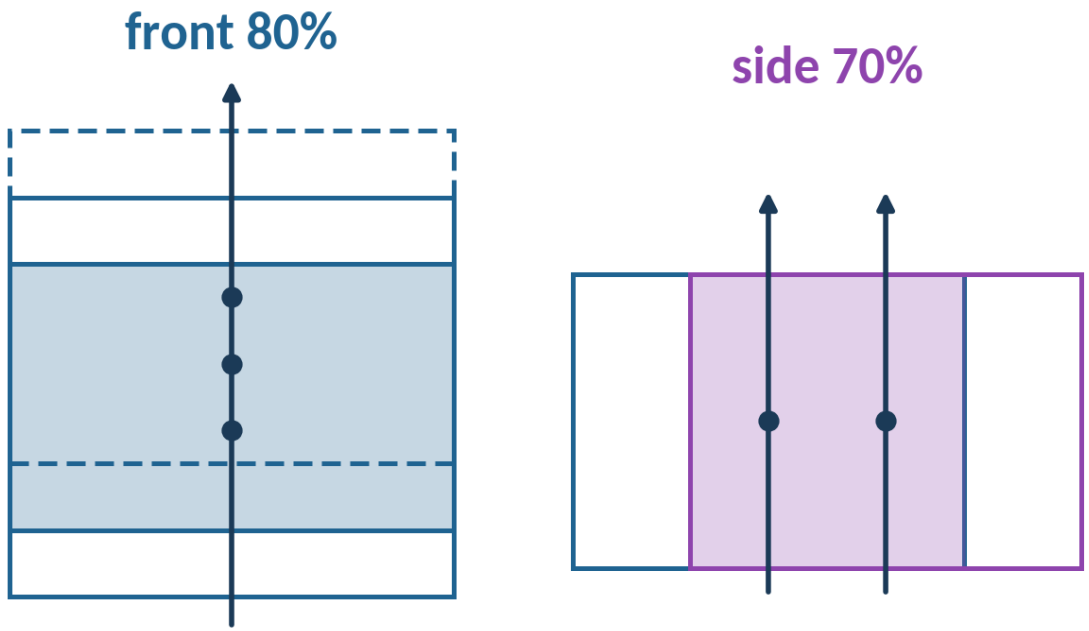
- **Photogrammetry:** the same idea as SfM, called aerial triangulation

Wikipedia: Structure from motion · Simultaneous localization and mapping · Epipolar geometry

Hartley, R., Zisserman, A. (2004): Multiple View Geometry in Computer Vision, 2nd ed. Cambridge University Press



Overlap



arrows: flight lines, dots: photo positions

- **Maps (2D):** 70–80 %
- **Trees, buildings:** ≈ 80–83 %
- **3D:** + cross grid, camera at 45°
- Front overlap is cheap, side overlap costs flight time

Wikipedia: Photogrammetry · Image stitching

Numbers: OpenDroneMap documentation, Flying tips (docs.opendronemap.org/flying)

Think first

Mavic 3M: the camera spec sheet

DJI Mavic 3M	RGB camera	multispectral camera
sensor	4/3 CMOS, 20 MP	1/2.8″ CMOS, 5 MP
image size	5280 × 3956 px	2592 × 1944 px
field of view	84° (diagonal)	73.91° (61.2° × 48.10°)
equivalent focal length	24 mm	25 mm
shortest photo interval	0.7 s (JPEG)	2 s (TIFF)
bands	RGB	G 560, R 650, RE 730, NIR 860 nm

Pause: which lines do you need for the pixel size on the ground? for the speed?

Wikipedia: Angle of view · 35 mm equivalent focal length

DJI Mavic 3M specifications, ag.dji.com/mavic-3-m/specs



Solution

Mavic 3M: the camera spec sheet

DJI Mavic 3M	RGB camera	multispectral camera
sensor	4/3 CMOS, 20 MP	1/2.8″ CMOS, 5 MP
image size	5280 × 3956 px	2592 × 1944 px
field of view	84° (diagonal)	73.91° (61.2° × 48.10°)
equivalent focal length	24 mm	25 mm
shortest photo interval	0.7 s (JPEG)	2 s (TIFF)
bands	RGB	G 560, R 650, RE 730, NIR 860 nm

Pixel size: image size + field of view · **speed:** photo interval

Not needed: equivalent focal length

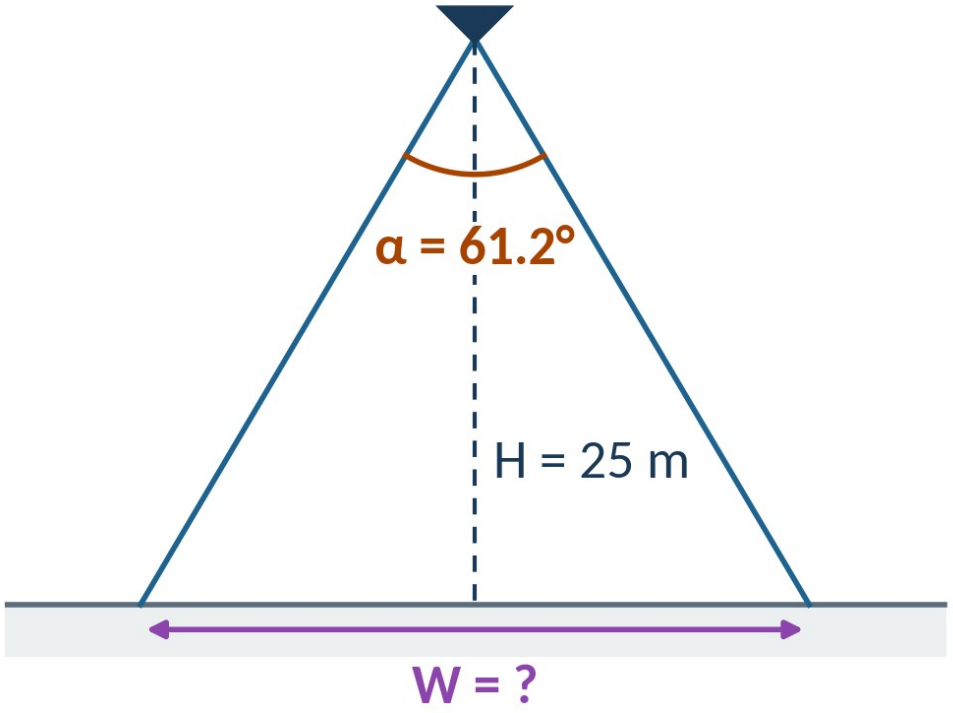
Wikipedia: Angle of view · 35 mm equivalent focal length

DJI Mavic 3M specifications, ag.dji.com/mavic-3-m/specs



Think first

Opening angle → footprint → pixel



Given

MS camera: 61.2° across 2592 px

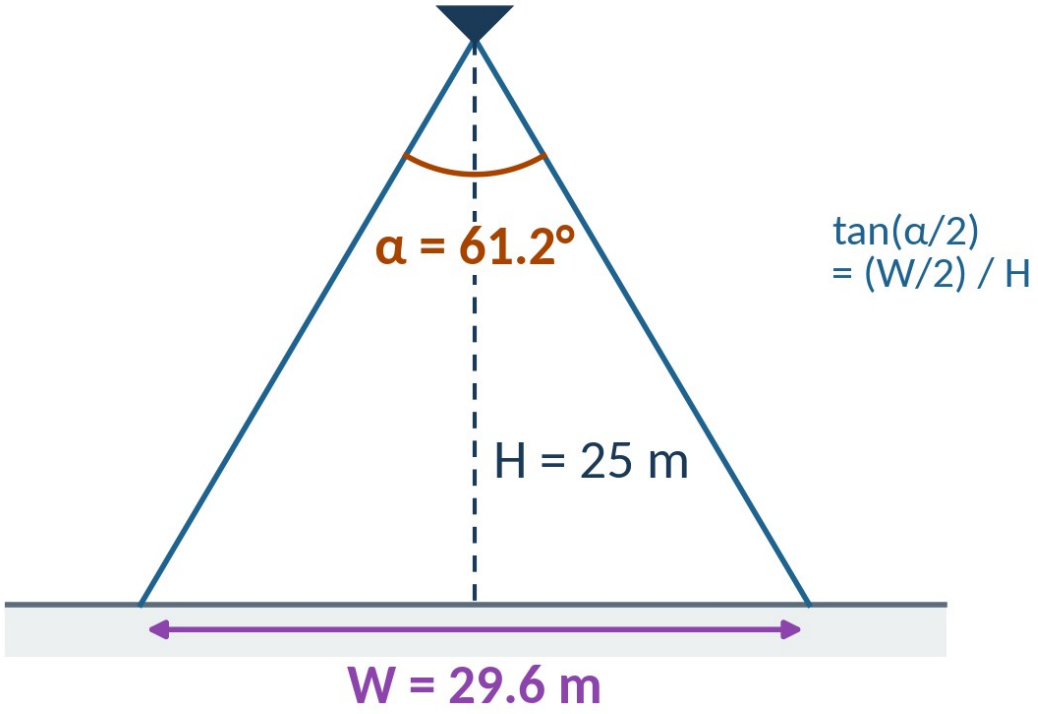
flying height H = 25 m

Pause: how wide is one photo on the ground? how large is one pixel?

Wikipedia: Angle of view · Ground sample distance

Solution

Opening angle → footprint → pixel



tan(α/2)
= (W/2) / H

Given

MS camera: 61.2° across 2592 px

flying height H = 25 m

1 **W = 2 · H · tan(α/2)**

= 2 · 25 m · tan(30.6°) = 29.6 m

2 **GSD = W / pixels**

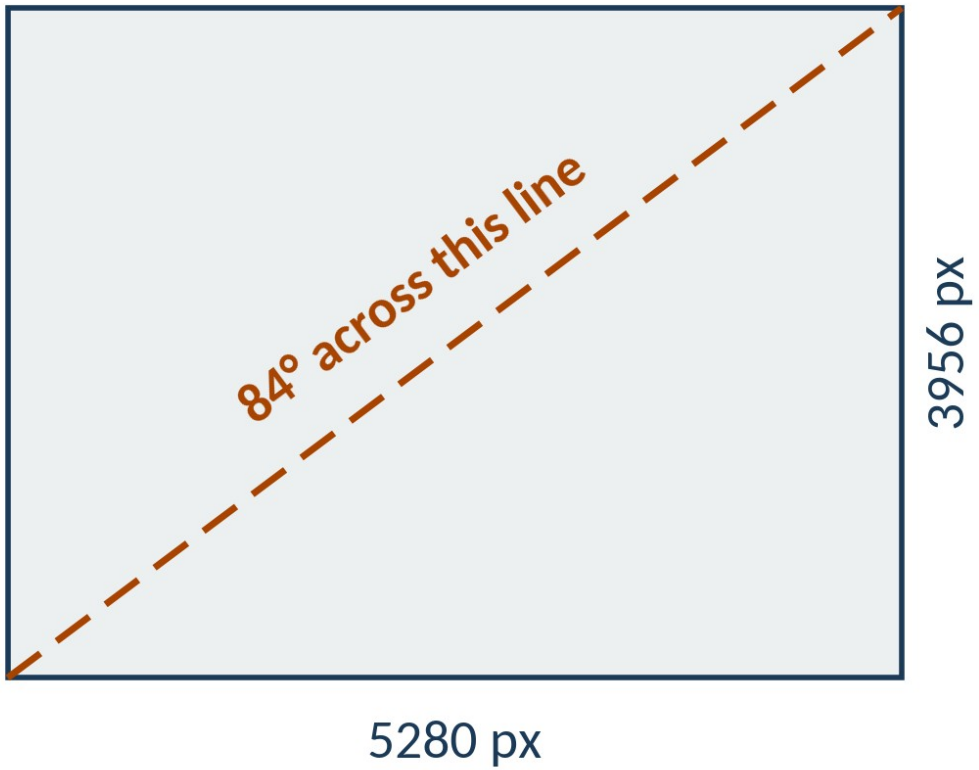
= 29.6 m / 2592 = 1.14 cm

Rule: GSD in cm ≈ H in m / 21.9

Wikipedia: Angle of view · Ground sample distance

Think first

RGB: the angle is across the diagonal



Given

RGB camera: 84° across the diagonal

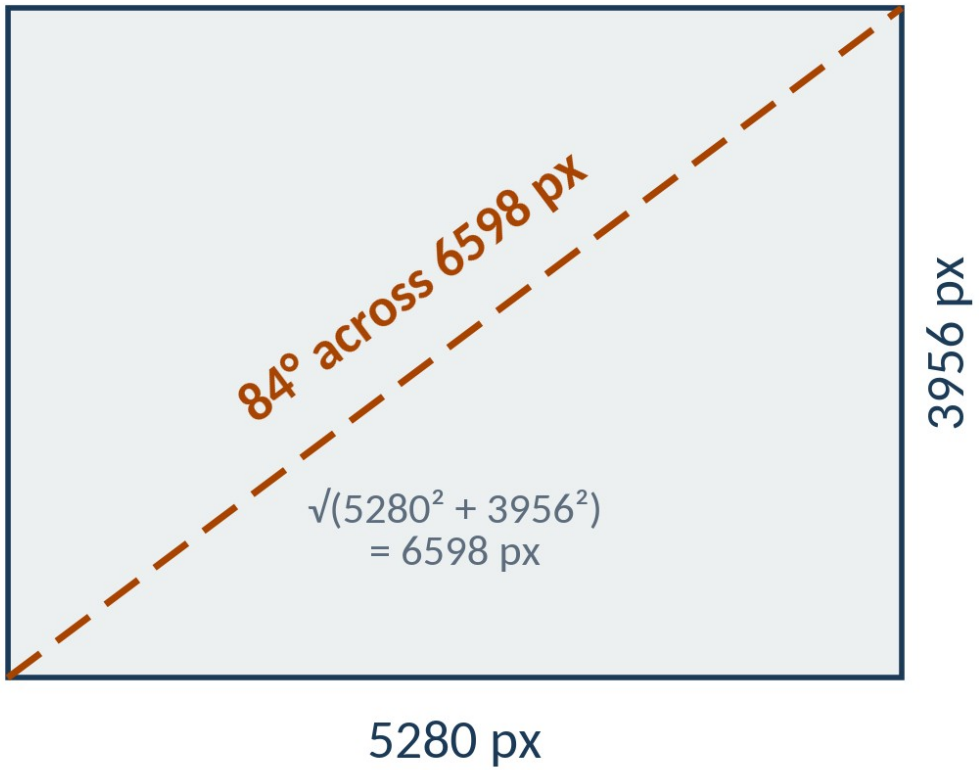
5280 × 3956 px, H = 25 m

Pause: the pixels are counted along x and y, the angle along the diagonal: how do you get the pixel size?

Wikipedia: Angle of view

Solution

RGB: the angle is across the diagonal



Given

RGB camera: 84° across the diagonal

5280 × 3956 px, H = 25 m

1 diagonal: $\sqrt{(5280^2 + 3956^2)} = 6598$ px

2 $D = 2 \cdot 25\ m \cdot \tan(42°) = 45.0$ m

3 **GSD = 45.0 m / 6598 = 0.68 cm**

photo: 36.0 × 27.0 m

Not: 84° × 5280/6598 = 67.2°

Wikipedia: Angle of view

Is the spec sheet consistent?

Self-study

check	computed	DJI states	agrees?
RGB: 24 mm equivalent → diagonal	84.1°	84°	yes
MS: 25 mm equivalent → diagonal	81.7°	73.91°	no
MS: 61.2° × 48.10° → diagonal	73.1°	73.91°	≈ 1 %

- **Rule:** use the angle that belongs to your pixel count
- Equivalent focal length: for photographers, not for mapping

DJI Mavic 3M specifications; full frame = 36 × 24 mm



Pixel size on the ground (GSD)

GSD = H × (width of the view on the ground per metre) / pixels

height	RGB pixel	MS pixel	RGB photo
25 m	0.7 cm	1.1 cm	36 × 27 m
40 m	1.1 cm	1.8 cm	58 × 43 m
60 m	1.6 cm	2.7 cm	86 × 65 m
120 m	3.3 cm	5.5 cm	173 × 130 m

Mavic 3M, GSD in cm: RGB ≈ H / 36.6, multispectral ≈ H / 21.9 (H in m)

Wikipedia: Ground sample distance

Camera: DJI Mavic 3M specifications (ag.dji.com/mavic-3-m/specs)



Sharp photos

blur = speed × exposure time (keep < ½ pixel)

speed	exposure	blur	at 25 m (RGB 0.68 cm)
2.2 m/s	1/1000 s	2.2 mm	0.3 px
7.7 m/s	1/1000 s	7.7 mm	1.1 px
7.7 m/s	1/250 s	30.8 mm	4.5 px

- Low light → longer exposure → fly slower
- **RGB: mechanical shutter** · MS: electronic

Wikipedia: Motion blur · Shutter speed · Rolling shutter

Light, weather, scene

- **Even light:** overcast is fine, broken clouds are not
- **November sun:** ≈ 23° at noon → shadows 2.4 × object height
- **Wind:** moving trees, blurred photos
- **Multispectral:** light sensor + reflectance panel before and after
- **Scene:** no water, glass, moving people or cars

Wikipedia: Solar zenith angle · Multispectral imaging



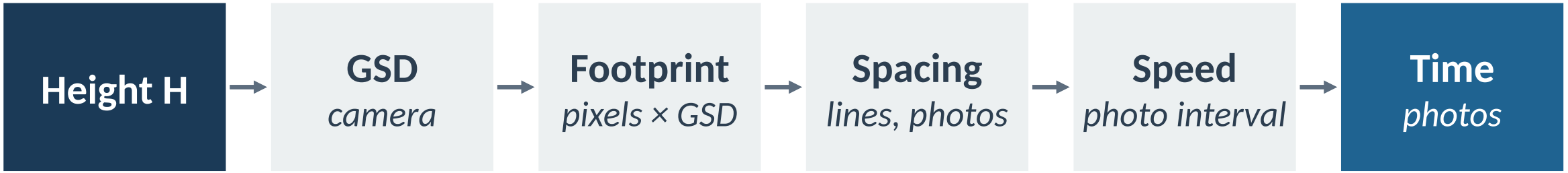
3 Plan

From flying height to flight lines

Photos → map | Control | Photos | **Plan**



From flying height to flight time

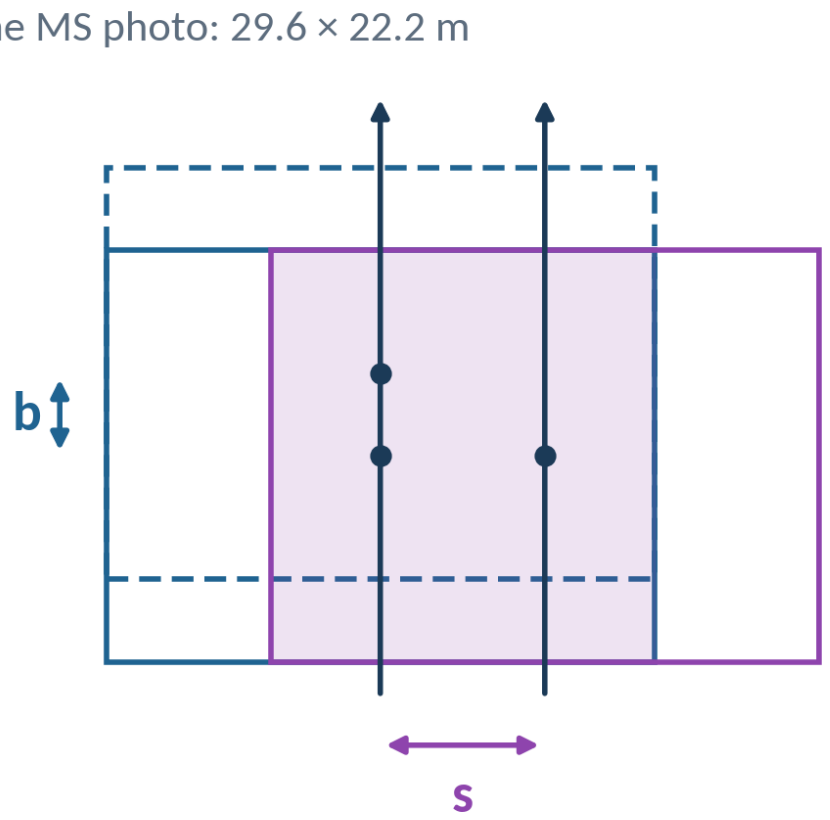


- **Line spacing** = footprint across × (1 − side overlap)
- **Photo spacing** = footprint along × (1 − front overlap)
- **Speed** = photo spacing / photo interval (RGB 0.7 s, MS 2 s)
- **Lower**: finer pixels, more lines, longer flight

Wikipedia: Ground sample distance · Photogrammetry

Think first

Line spacing



Given

MS photo across the line: 29.6 m

side overlap 70 %

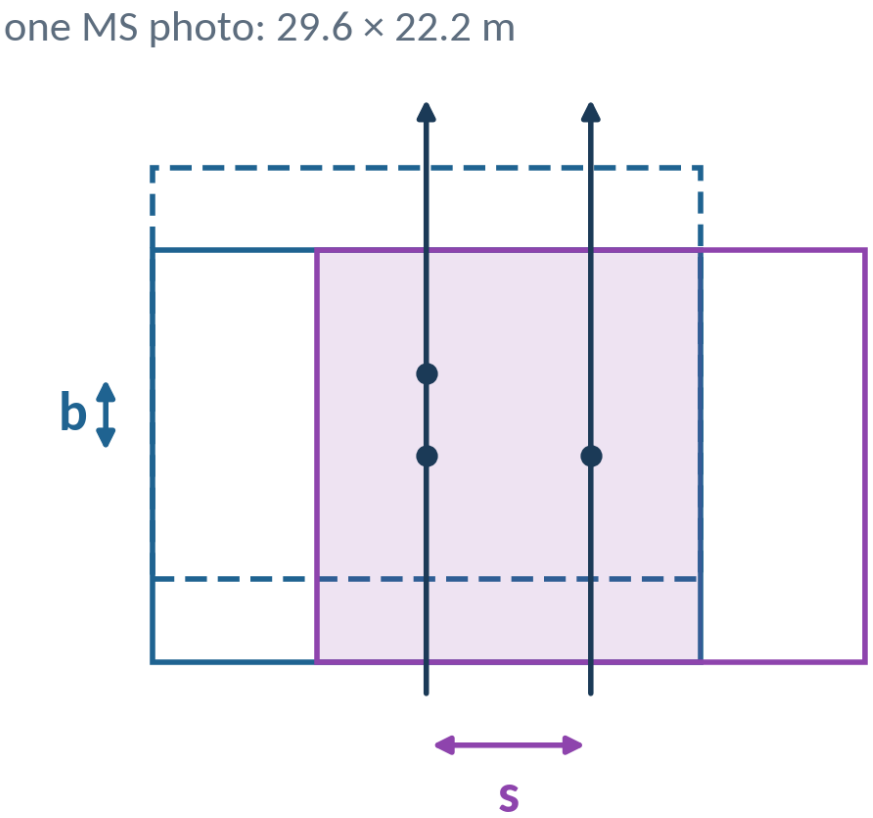
area ≈ 48 m across the lines

Pause: how far apart are the flight lines (s)? how many lines?

Wikipedia: Photogrammetry

Solution

Line spacing



Given

MS photo across the line: 29.6 m

side overlap 70 %

area ≈ 48 m across the lines

1 **s = across · (1 – side overlap)**

= 29.6 m · 0.3 = 8.9 m

2 lines = ⌈48 / 8.9⌉ = 6

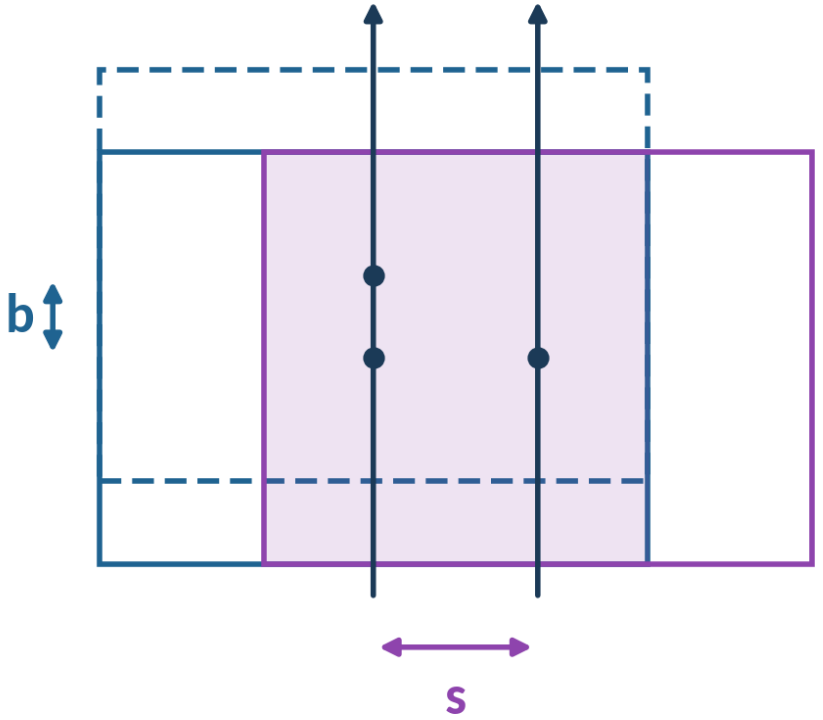
RGB alone: 10.8 m → 5 lines

Wikipedia: Photogrammetry

Think first

Photo spacing and speed

one MS photo: 29.6 × 22.2 m



Given

MS photo along the line: 22.2 m

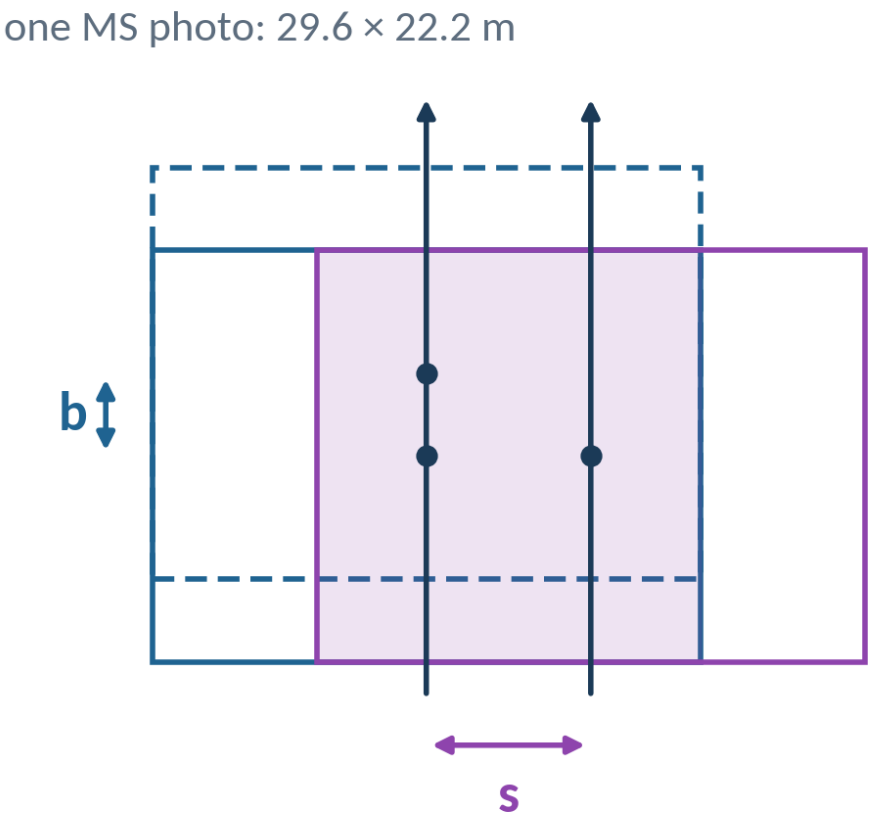
front overlap 80 %

MS: one photo every 2 s

Pause: how far apart are the photos (b)? how fast may the drone fly?

Solution

Photo spacing and speed



Given

MS photo along the line: 22.2 m

front overlap 80 %

MS: one photo every 2 s

1 **b = along · (1 − front overlap)**

= 22.2 m · 0.2 = 4.4 m

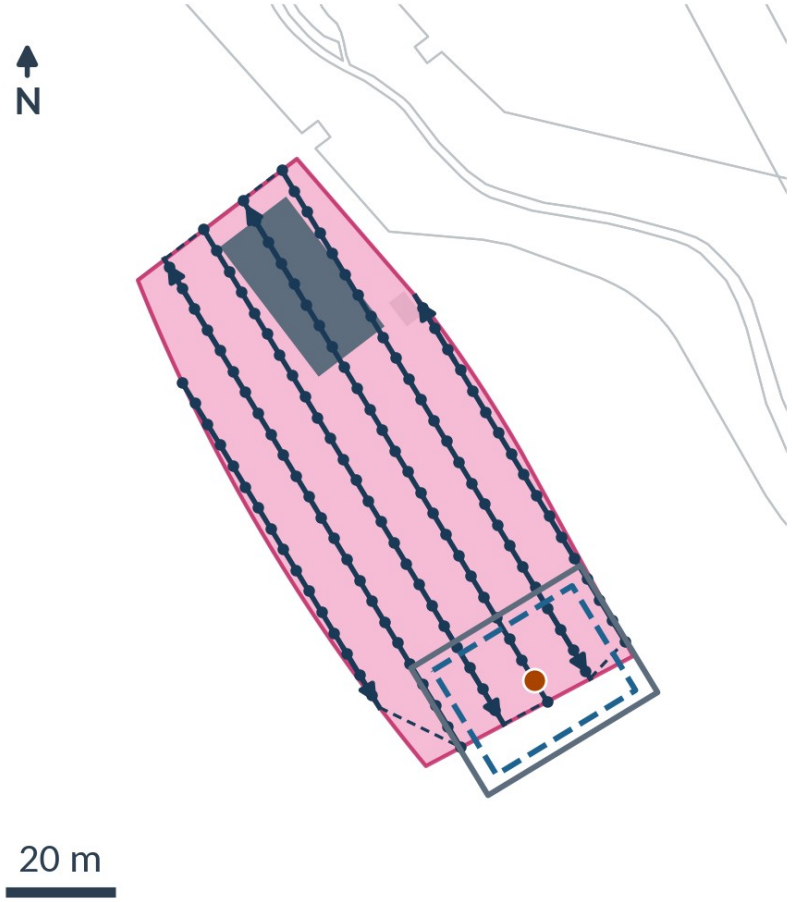
2 **v = b / interval = 4.4 m / 2 s = 2.2 m/s**

RGB alone: 5.4 m / 0.7 s = 7.7 m/s

→ the MS camera decides

Think first

Flight time and number of photos



Given

6 lines, path 618 m (lines + turns)

v = 2.2 m/s, ≈ 10 s per turn

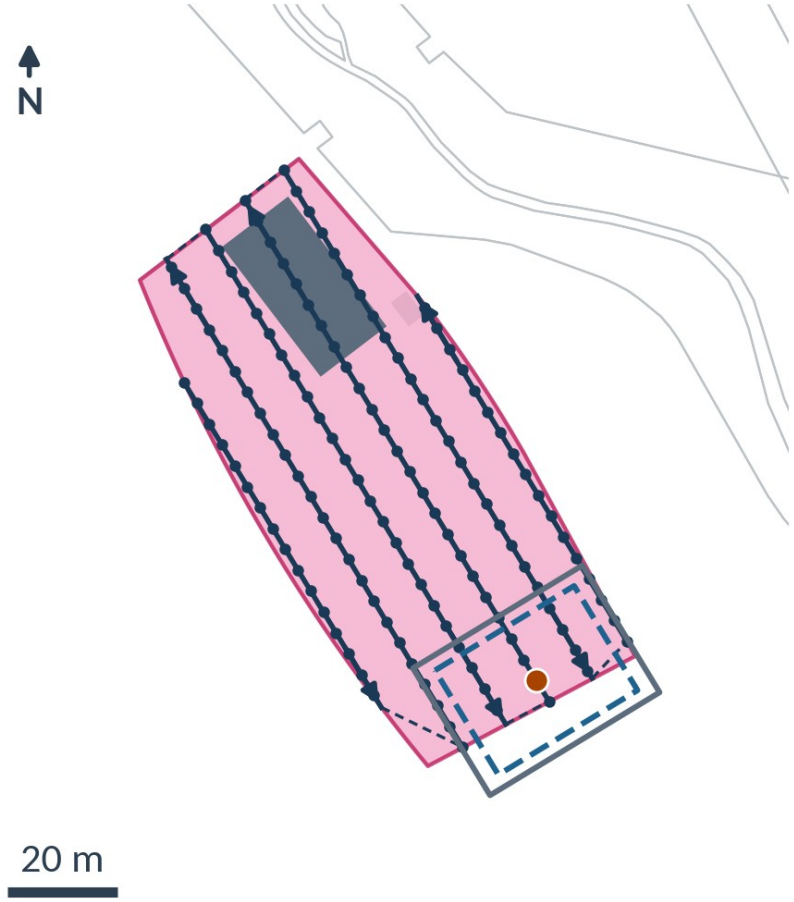
photo every 4.4 m

Pause: how long is the flight? how many photo positions and image files?

Wikipedia: Photogrammetry

Solution

Flight time and number of photos



Given

6 lines, path 618 m (lines + turns)

v = 2.2 m/s, ≈ 10 s per turn

photo every 4.4 m

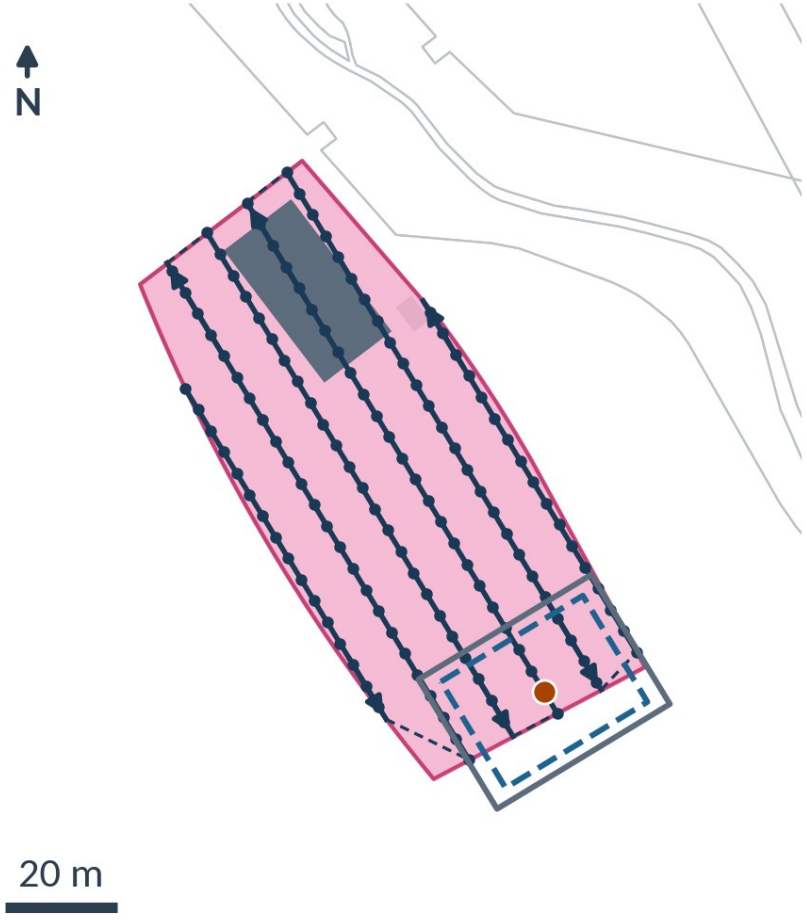
1 t = 618 m / 2.2 m/s + 5 × 10 s
= 279 s + 50 s ≈ 5.5 min

2 per line ⌊length / b⌋ + 1 → 131 positions

3 × 5 images (RGB + 4 bands) ≈ 655 files

Wikipedia: Photogrammetry

The Green Fab Lab plan



all cameras, 25 m	
pixel RGB / MS	0.7 / 1.1 cm
lines, spacing	6 × 8.9 m
photo every	4.4 m
speed	2.2 m/s
photo positions	131
flight time	≈ 5 min

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; plan: R. Becker (common.mission); camera: DJI Mavic 3M specs

All cameras: the multispectral camera decides

at 25 m	RGB	multispectral
pixel	0.68 cm	1.14 cm
photo on the ground	36 × 27 m	30 × 22 m
line spacing (70 %)	10.8 m	8.9 m
photo spacing (80 %)	5.4 m	4.4 m
photo interval	0.7 s	2 s
max. speed	7.7 m/s	2.2 m/s

Pause: what changes if we fly at 40 m?



Before take-off

Rules

Registered operator, certified pilot

Visual line of sight, ≤ 120 m

Not over uninvolved people

Geo-zones: dipul.de

On site

Public park: watch the paths

Wind, rain, light

Batteries, home point

RTK fix before take-off

Wikipedia: Regulation of unmanned aerial vehicles

Rules: Regulation (EU) 2019/947 (open category); geo-zones: dipul.de. The pilot in command decides.



Plan in Python, fly with DJI Pilot 2



- You compute the numbers; the app flies the lines
- Same height and overlap in both
- **Homework:** your plan for the Green Fab Lab

Wikipedia: Keyhole Markup Language

Summary

- GCPs fix the model, check points test it – never mix them
- Drone RTK: GCPs optional, check points still needed
- Every point in ≥ 3–5 photos: 70–80 % overlap, texture, a still scene
- Height sets pixel size, lines, photos and time
- **All cameras: the multispectral camera decides**



Sources and further reading (1/3)

From photos to a map: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Ground control and check points: Ground control point · Accuracy and precision · Root mean square deviation

How many, and where?: Ground control point · Bundle adjustment

Georeferencing: two questions: Real-time kinematic positioning · Ground control point · Orthophoto

What must my map fit?: Image registration · Accuracy and precision

Our choice at the Green Fab Lab: Real-time kinematic positioning

Targets you can find in the photos: Fiducial marker · Ground sample distance

Our base mark at the Green Fab Lab: Survey marker · Tribrach

Structure from motion: Structure from motion · Feature (computer vision) · Bundle adjustment

Bundle adjustment: make the rays meet: Bundle adjustment · Triangulation (computer vision)

Reprojection error: Reprojection error · Levenberg–Marquardt algorithm

A model from photos alone floats: Similarity (geometry) · Aerial triangulation



Sources and further reading (2/3)

SfM, SLAM, multiple view geometry: Structure from motion · Simultaneous localization and mapping · Epipolar geometry

Overlap: Photogrammetry · Image stitching

Mavic 3M: the camera spec sheet: Angle of view · 35 mm equivalent focal length

Opening angle → footprint → pixel: Angle of view · Ground sample distance

RGB: the angle is across the diagonal: Angle of view

Pixel size on the ground (GSD): Ground sample distance

Sharp photos: Motion blur · Shutter speed · Rolling shutter

Light, weather, scene: Solar zenith angle · Multispectral imaging

From flying height to flight time: Ground sample distance · Photogrammetry

Line spacing: Photogrammetry

Flight time and number of photos: Photogrammetry

Before take-off: Regulation of unmanned aerial vehicles



Sources and further reading (3/3)

Plan in Python, fly with DJI Pilot 2: Keyhole Markup Language

Text distilled from Wikipedia (CC BY-SA 4.0), the OpenDroneMap documentation (docs.opendronemap.org), the DJI Mavic 3M specifications and the ASPRS Positional Accuracy Standards, Edition 2 (2023). Maps: ALKIS and height benchmarks, Geobasis NRW (dl-de/zero-2-0). Figures: R. Becker (make_slides.py), CC BY-SA 4.0.



Licence and how to cite

Cite as: Becker, R. (2026): geo1260 Drone mapping. opengeo teaching material, Earth Observation Lab (EOLab-HSRW), Hochschule Rhein-Waal, eolab.de. CC BY-SA 4.0.

- **Licence:** CC BY-SA 4.0 – creativecommons.org/licenses/by-sa/4.0. You may share and adapt this material if you credit the author and keep the licence.
- **Not covered:** the logos of Hochschule Rhein-Waal and EOLab-HSRW, and third-party material with its own credit on the slide (ALKIS and height benchmarks: Geobasis NRW, dl-de/zero-2-0; texts after Wikipedia: CC BY-SA).
- **Contact:** EOLab-HSRW, Hochschule Rhein-Waal, Kamp-Lintfort · eolab.de